

Mohamed Aziz Tlili

AI & Full Stack Software Engineer

Tunis, Tunisia | medaziztlilipro@gmail.com | +216 94 157 831

linkedin.com/in/med-aziz-tlili | github.com/med-tlili

PROFESSIONAL SUMMARY

AI and Full Stack Software Engineer specializing in computer vision and deep learning, with hands-on experience building real-time detection and tracking systems (YOLOv8, MediaPipe, Kalman filtering) and shipping full-stack applications with FastAPI, React, and Flutter. Currently completing a final-year engineering project developing the computer vision and AI coaching pipeline for an intelligent sports analytics platform.

EDUCATION

Institut Supérieur d'Informatique et de Gestion de Kairouan (ISIGK)

Kairouan, Tunisia

Bachelor's Degree (Licence) in Computer Science — Software Engineering, Focus on Deep Learning & Computer Vision | Expected 2026

EXPERIENCE

Software Engineering Intern — ULTIMA ICD

2024 (4 months)

- Developed and deployed AI-powered applications using Python, TensorFlow, and OpenCV.
- Built full-stack web interfaces with React, TypeScript, and Tailwind CSS.
- Implemented computer vision pipelines for real-time object detection and tracking.

PROJECTS

SUMMA v3 — Intelligent Padel Court Management & Analysis Platform

2024 – 2026

Final-Year Engineering Project (PFE), in partnership with ULTIMA ICD | Team of 2 | Computer Vision, AI, Full Stack

Owned the computer vision and AI scope of a two-person team project; teammate delivered the Flutter mobile app and Supabase backend.

- Built an end-to-end computer vision pipeline transforming raw padel match footage into broadcast-style analysis, including player and ball detection with YOLOv8 and Kalman filtering.
- Developed a VAR (Video Assistant Referee) highlight engine that automatically detects key match events and generates zoomed, slow-motion replay clips.
- Designed an AI coaching module using MediaPipe pose estimation to extract player keypoints and generate biomechanical feedback on shot technique.
- Implemented automated shot classification (Smash, Vibora, Bandeja) using biomechanical rule-based logic.
- Collaborated on system architecture and integration with a teammate building the Flutter admin/player apps and Supabase backend.

Tech: Python, PyTorch, YOLOv8, OpenCV, MediaPipe, Kalman Filter, FastAPI, Flutter, Supabase, Docker

TempGuard — IoT Temperature Monitoring System

2024

Mobile Application • IoT

- Designed and built a real-time temperature monitoring mobile application for multi-room facilities.
- Implemented a live dashboard with color-coded room status (OK / Warning / Critical) and threshold-based push notifications.
- Built historical data visualization with date-range filtering and an alert acknowledgment flow.

Tech: Flutter, Dart, Provider State Management, fl_chart

Chess Game

2024

Web Application • Game Development

- Developed a web-based chess game supporting local multiplayer and AI opponent modes.
- Implemented game logic, move validation, and an AI engine for computer opponent play.

Tech: React, TypeScript

Computer Vision Modules

2023 – 2024

Machine Learning • Computer Vision

- Built a hand detection system that triggers PC actions based on recognized gestures (e.g., clap to trigger an event).
- Developed a face detection, recognition, and expression analysis pipeline using deep learning models.
- Trained ML prediction models to forecast tournament winners from historical data.

Tech: Python, OpenCV, MediaPipe, TensorFlow, Scikit-learn

SKILLS

Languages	Python, TypeScript, JavaScript, Kotlin, Dart, SQL
AI / ML	PyTorch, TensorFlow, YOLOv8, MediaPipe, LangChain, HuggingFace, OpenAI API
Computer Vision	OpenCV, YOLOv8, ByteTrack, Kalman Filter, Pose Estimation, Object Detection & Tracking
Frontend	React, Next.js, TypeScript, Tailwind CSS, HTML/CSS
Backend	FastAPI, Node.js, Express, Django, REST APIs, GraphQL
Mobile	Flutter, Kotlin, React Native
Database	PostgreSQL, MongoDB, Redis, SQLite
DevOps	Docker, GitHub Actions, Linux, Git
Tools	VS Code, Figma, Jupyter, Google Colab, Roboflow, CVAT